QUBO Formulation: Minimum Graph Bisection Problem

Minimum Graph Bisection Problem

Problem Statement. Given an undirected weighted graph $G = (V, E)$ with an even number of vertices $N = |V|$ and non-negative edge weights w_{ij} for each $(i, j) \in E$, partition the vertex set V into two disjoint subsets V_A and V_B such that $|V_A| = |V_B| = N/2$. The objective is to minimize the total weight of edges crossing between the two partitions, i.e., minimize $\sum_{(i,j)\in E} w_{ij}\mathbb{I}(i \in V_A, j \in V_B \text{ or } i \in V_B, j \in V_A)$.

Decision Variables. For each vertex $i \in V$, introduce a binary variable $x_i \in \{0, 1\}$. The assignment $x_i = 0$ indicates that vertex i belongs to partition V_A, while $x_i = 1$ indicates membership in partition V_B. The set of all variables is $\{x_i\}_{i\in V}$.

QUBO Derivation. Step 1 — Objective Formulation. The cut weight can be expressed using the binary variables as $\sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2$, since $(x_i - x_j)^2 = 1$ when $x_i \neq x_j$ and 0 otherwise. For binary variables, the identity $(x_i - x_j)^2 = x_i + x_j - 2x_ix_j$ holds. Thus, the minimization objective is:

$$\min \sum_{(i,j)\in E} w_{ij}(x_i + x_j - 2x_ix_j). \tag{1}$$

Step 2 — Constraint Formulation. The requirement that both partitions have equal size is enforced by the equality constraint:

$$\sum_{i\in V} x_i = \frac{N}{2}. \tag{2}$$

Step 3 — Penalty Term Expansion. We convert the equality constraint into a quadratic penalty term with parameter $\lambda > 0$:

$$P(x) = \lambda \left(\sum_{i\in V} x_i - \frac{N}{2} \right)^2. \tag{3}$$

Expanding the square yields:

$$P(x) = \lambda \left(\sum_{i \in V} x_i^2 - N \sum_{i \in V} x_i + \frac{N^2}{4} + \sum_{i \neq j} x_i x_j \right). \tag{4}$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, and noting that $\sum_{i \neq j} x_i x_j = 2 \sum_{i<j} x_i x_j$, we obtain:

$$P(x) = \lambda \left((1 - N) \sum_{i \in V} x_i + 2 \sum_{i<j} x_i x_j + \frac{N^2}{4} \right). \tag{5}$$

Step 4 — Combined Hamiltonian. Adding the objective and penalty terms gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = \sum_{(i,j) \in E} w_{ij}(x_i + x_j - 2x_i x_j) + \lambda \left((1 - N) \sum_{i \in V} x_i + 2 \sum_{i<j} x_i x_j + \frac{N^2}{4} \right). \tag{6}$$

Grouping linear and quadratic terms, and defining the weighted degree $\deg_w(i) = \sum_{j:(i,j) \in E} w_{ij}$, we rewrite $H(x)$ as:

$$H(x) = \sum_{i \in V} \left(\deg_w(i) + \lambda(1 - N) \right) x_i + \sum_{(i,j) \in E} (-2w_{ij}) x_i x_j + \sum_{i<j} (2\lambda) x_i x_j + \frac{\lambda N^2}{4}. \tag{7}$$

Step 5 — Q Matrix and Offset. The Hamiltonian takes the standard QUBO form $H(x) = x^T Q x + c$. The matrix entries and scalar offset are:

$$Q_{i,i} = \deg_w(i) + \lambda(1 - N), \quad \forall i \in V, \tag{8}$$

$$Q_{i,j} = \begin{cases} -2w_{ij} + 2\lambda & \text{if } (i,j) \in E, \\ 2\lambda & \text{if } i \neq j \text{ and } (i,j) \notin E, \\ 0 & \text{otherwise,} \end{cases} \tag{9}$$

$$c = \frac{\lambda N^2}{4}. \tag{10}$$

Penalty Weight Justification. To guarantee that the global minimum of $H(x)$ over $\{0,1\}^N$ corresponds to a feasible bisection, the penalty parameter λ must dominate any potential reduction in the cut weight achievable by violating the balance constraint. Flipping a single variable changes the constraint violation by exactly 1, increasing the penalty by λ. The maximum possible decrease in the objective function from a single bit flip is bounded by the maximum weighted degree $\Delta_w = \max_{i \in V} \deg_w(i)$. Setting $\lambda > \frac{\Delta_w}{N+1}$ ensures that the energy cost of a unit constraint violation strictly exceeds any feasible objective improvement, thereby forcing the optimizer toward feasible configurations.

Correctness Argument. Minimizing $H(x)$ over binary vectors recovers the optimal bisection because (i) all feasible assignments satisfying $\sum_{i \in V} x_i = N/2$ incur zero penalty, reducing $H(x)$ exactly to the original cut weight objective; and (ii) any infeasible assignment incurs a penalty strictly greater than the maximum possible decrease in the cut weight from any single variable flip. Consequently, the global minimum of $H(x)$ must occur at a feasible point, and among feasible points, $H(x)$ is identical to the cut weight. Thus, the minimizer of $H(x)$ corresponds precisely to the minimum-weight balanced partition.

Illustrative Example. Consider a path graph with $N = 4$ vertices $V = \{0, 1, 2, 3\}$ and edges $E = \{(0, 1), (1, 2), (2, 3)\}$ with weights $w_{01} = 1$, $w_{12} = 2$, $w_{23} = 1$. The weighted degrees are $\deg_w(0) = 1$, $\deg_w(1) = 3$, $\deg_w(2) = 3$, $\deg_w(3) = 1$. We choose $\lambda = 1$, which satisfies $\lambda > \Delta_w/(N + 1) = 3/5$. Substituting these values into the general formulas yields the concrete QUBO parameters:

$$Q_{0,0} = 1 + 1(1 - 4) = -2, \qquad Q_{1,1} = 3 + 1(1 - 4) = 0, \tag{11}$$

$$Q_{2,2} = 3 + 1(1 - 4) = 0, \qquad Q_{3,3} = 1 + 1(1 - 4) = -2, \tag{12}$$

$$Q_{0,1} = -2(1) + 2(1) = 0, \qquad Q_{1,2} = -2(2) + 2(1) = -2, \tag{13}$$

$$Q_{2,3} = -2(1) + 2(1) = 0, \qquad Q_{0,2} = 2(1) = 2, \tag{14}$$

$$Q_{0,3} = 2(1) = 2, \qquad Q_{1,3} = 2(1) = 2, \tag{15}$$

$$c = \frac{1 \cdot 4^2}{4} = 4. \tag{16}$$

The resulting Hamiltonian is:

$$H(x) = -2x_0 - 2x_3 - 2x_1x_2 + 2x_0x_2 + 2x_0x_3 + 2x_1x_3 + 4. \tag{17}$$

Evaluating $H(x)$ on the balanced assignment $x = (1, 1, 0, 0)$ (partition $\{0, 1\}$ vs $\{2, 3\}$) gives $H(1, 1, 0, 0) = -2(1) - 2(0) - 2(0) + 0 + 0 + 0 + 4 = 2$. This matches the cut weight of the partition, which consists solely of the edge $(1, 2)$ with weight 2. The global minimum of $H(x)$ over all balanced assignments is 2, confirming that $\{0, 1\}$ and $\{2, 3\}$ form the minimum cut bisection.